

Distributed MEMS Analog Phase Shifter With Enhanced Tuning

Greg McFeetors, *Student Member, IEEE*, and Michal Okoniewski, *Senior Member, IEEE*

Abstract—The design, fabrication, and measurement of a tunable microwave phase shifter is described. The phase shifter combines two techniques: a distributed capacitance transmission line phase shifter, and a large tuning range radio frequency (RF) microelectromechanical system (MEMS) capacitor. The resulting device is a large bandwidth, continuously tunable, low-loss phase shifter, with state-of-the-art performance. Measurements indicate analog tuning of 170° phase shift per dB loss is possible at 40 GHz, with a 538° phase shift per centimeter. The structure is realized with high-Q MEMS varactors, capable of tuning $C_{\max}/C_{\min} = 3.4$. To our knowledge, this presents the lowest loss analog millimeter wave phase shifter performance to date.

Index Terms—Distributed microelectromechanical transmission line (DMTL), microelectromechanical system (MEMS), phase shifter, varactor.

I. INTRODUCTION

MICROWAVE phase shifters find uses in many communications circuits, and for several applications, such as phased antenna arrays, phase shifter loss directly impacts the signal's dynamic range. Moreover, the large size and high cost of ferrite based phase shifters that are dominant in antenna arrays prevent widespread use of phased-arrays in public communication and related industries. Microelectromechanical system (MEMS) circuits have the potential to be inexpensive, have low loss, and have high quality phase-shifting capability. They can also be integrated circuit (IC) compatible, and are thus harmonized with the quest of achieving integrated and ever smaller radio frequency (RF) front ends.

The most abundant MEMS phase shifter designs use MEMS switches to choose between various signal path lengths [1]. However, switched-path type phase shifters are inherently digital, and quickly become cumbersome if a large number of phase states are desired. Devices such as phased antenna arrays often require high resolution in phase control, which would lead to a large and lossy digital phase shifter system.

An alternative phase shifter design uses MEMS tunable capacitors to periodically load a high impedance ($>50 \Omega$) transmission line [2], thus allowing for phase velocity and, consequently, phase shift to be continuously varied [3]. Such a phase shifter design has been well documented by Barker [4], and referred to as a distributed MEMS transmission line (DMTL). We

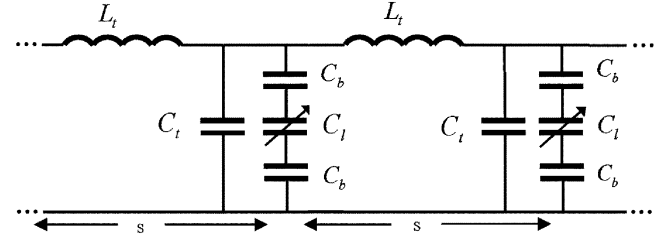


Fig. 1. Periodically loaded transmission line equivalent circuit. C_b represents electrode capacitance, which is explained in Section III.

present a device that belongs to the same family. We have, however, developed it with a custom fabrication process to minimize insertion loss, and with high tuning RF capacitors to maximize achievable phase shift. The resulting device has demonstrated over three times higher phase shift per dB of loss than previously reported in literature for a continuously variable MEMS phase shifter.

II. THEORY

A transmission line is periodically loaded with shunt capacitors, as shown in Fig. 1. Here, the line's per unit length capacitance, C_t , and inductance, L_t , are complimented by the periodically spaced lumped shunt capacitance, C_l . With an electrically small separation, s , these lumped capacitors appear to be distributed along the line. The periodic capacitors enhance the natural capacitance of the transmission line structure, altering its characteristic impedance, phase velocity, and hence phase delay across the line. The phase delay is then dependant on C_l as

$$\Phi = \beta l = 2\pi f l \sqrt{L_t \left(C_t + \frac{C_l}{s} \right)}. \quad (1)$$

If C_l is variable, (1) illustrates the phase shifting operation of the device.

If C_l is realized using MEMS techniques, the distributed capacitance can be tuned, such as by electrostatic actuation. This implementation, shown in Fig. 2, utilizes MEMS bridge shunt capacitors. For a simple bridge, dc voltage between signal and ground can be used to cause electrostatic actuation. However, a more sophisticated actuation scheme is used in our implementation (see Section III).

Taking into account the reflections from the periodic loading of the line, the characteristic impedance of the loaded transmission line is

$$Z_0 = \sqrt{\frac{sL_t}{sC_t + C_l}} \sqrt{1 - \frac{\omega}{\omega_B}} \quad (2)$$

where ω_B is the Bragg frequency [5].

Manuscript received June 24, 2005; revised September 27, 2005. This work was supported by the National Science and Engineering Research Council of Canada, by TRILabs, and by the University of Alberta Nanofab. The review of this letter was arranged by Associate Editor A. Weisshaar.

The authors are with the Department of Electrical and Computer Engineering, University of Calgary, Calgary, AB T2N 1N4, Canada (e-mail: okoniewski@ucalgary.ca).

Digital Object Identifier 10.1109/LMWC.2005.861350

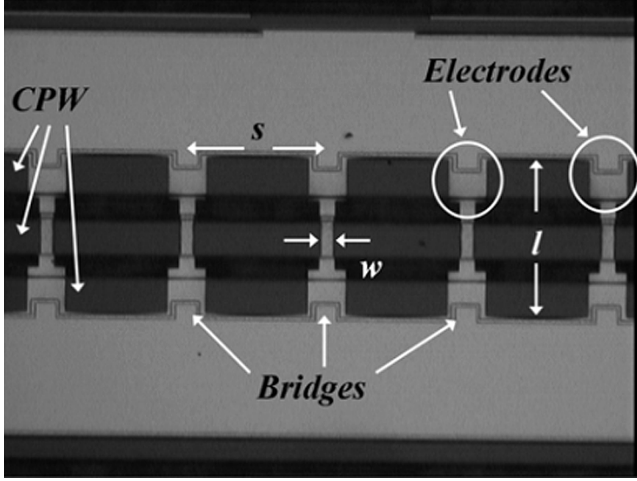


Fig. 2. Photograph of a portion of the DMTL phase shifter.

Based on the equivalent transmission line circuit, the phase shift can be predicted as

$$\Delta\phi = f\sqrt{L_t} \left(\sqrt{C_t + \frac{C_{t_{up}}}{s}} - \sqrt{C_t + \frac{C_{t_{down}}}{s}} \right) 360^\circ/\text{m}. \quad (3)$$

This simple analysis, based on the lumped element transmission line equivalent circuit, proves to be an accurate design tool, as illustrated in Section V.

III. DESIGN

A CPW line with an unloaded $Z_0 = 93 \Omega$ is used as the base transmission line. The line has a $82\text{-}\mu\text{m}$ wide center conductor and $76\text{-}\mu\text{m}$ gaps. This line is then loaded with 17 MEMS bridges, spaced $s = 300 \mu\text{m}$ apart to form a distributed loaded transmission line of 5.3-mm length. The fabricated device is shown in Fig. 2. Each MEMS bridge is 950-nm high at center (h_c), $28\text{-}\mu\text{m}$ wide (w), and $400\text{-}\mu\text{m}$ long (l). The MEMS bridges load this line to 67Ω with 0 V bias, 50Ω with a half deflection bias of 12 V and 44Ω with maximum bridge deflection at 16 V .

Tuning range before pull-in is limited in fixed-fixed type MEMS capacitor structures. A flat bridge can only be deflected by a maximum of approximately 33% of its initial height when force is applied at the center [6], resulting in a maximum 50% capacitance tuning range, or a tuning factor 1.5:1.

A method for far greater tuning has been proposed by Zou [7] using a two-layered bridge, as shown in Fig. 3. This architecture avoids the pull-in phenomena by separating the capacitance plates (h_c) from the electrostatic electrodes (h_e). Therefore, the pull-in deflection at the electrodes corresponds to a larger deflection ratio at the capacitor. With planar beams, the tuning factor of this enhanced capacitor can be defined as R:1, where

$$R \simeq \frac{\epsilon_r h_c + t_0}{\epsilon_r h_{c_{\min}} + t_0} \quad (4)$$

and $h_{c_{\min}}$ is the minimum height of the capacitor portion of the bridge. Capacitance tuning of this design is limited only by fabrication processing constraints, oxide thickness, and bridge planarity.

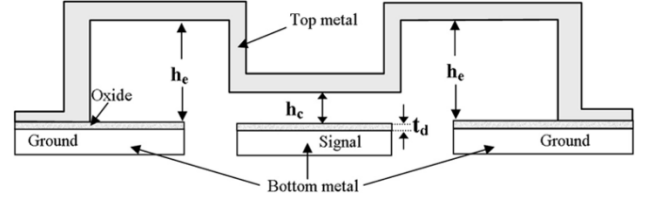


Fig. 3. High tuning two-level bridge capacitor profile.

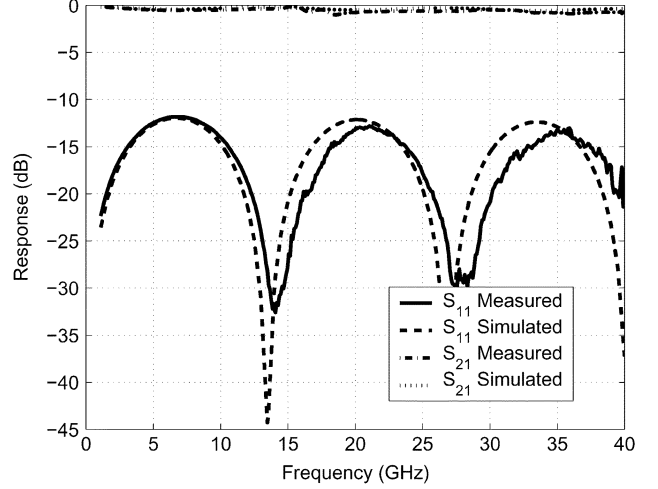

 Fig. 4. Measured and simulated S -parameters of the DMTL at 0 V bias.

Fig. 3 also shows the adaptation of this design to a CPW shunt capacitor. The center conductor forms the bottom plate of the capacitor, while the ground conductors are used as the bottom electrodes for electrostatic actuation. A thin layer, t_d , of oxide is used to insulate the top and bottom electrodes. The top metal is also shaped to form a large capacitance ($>50 \text{ pF}$) to the bottom ground conductor, C_b (Fig. 1).

Previously, this capacitor design has been proposed and fabricated using polysilicon MEMS processes, which leads to high loss at RF. In contrast, we have designed a custom RF process that allowed us to fabricate a high quality microwave phase shifter.

IV. FABRICATION

Copper has very desirable properties for use in a DMTL structure. Its high bulk conductivity ($\sigma = 60.8 \times 10^6$) gives low signal loss and thus makes it suitable for the base CPW layer. Aluminum is also a good RF metal ($\sigma = 37.6 \times 10^6$), has excellent adhesion, and can be deposited with low stress ($<50 \text{ MPa}$) and good planarity, and it is therefore selected for the released bridge layer.

The copper CPW lines were sputtered with adhesion and oxidation layers and patterned onto a fused silica substrate ($\epsilon_r = 3.79$, $\tan\delta = 2 \times 10^{-5}$). A 280-nm PECVD insulating oxide layer was then patterned onto the center conductor of the CPW line to avoid dc a short of the bridge to the bottom actuating electrode. The same oxide is used on the CPW ground planes to create a large capacitance C_b between the ground lines and the bridge. This oxide must be carefully processed to avoid dielectric breakdown during actuation.

Two sacrificial layers were then deposited and patterned to produce dimples and anchors with suitable sidewalls. The two consecutive sacrificial layers allows for the bridge height to be

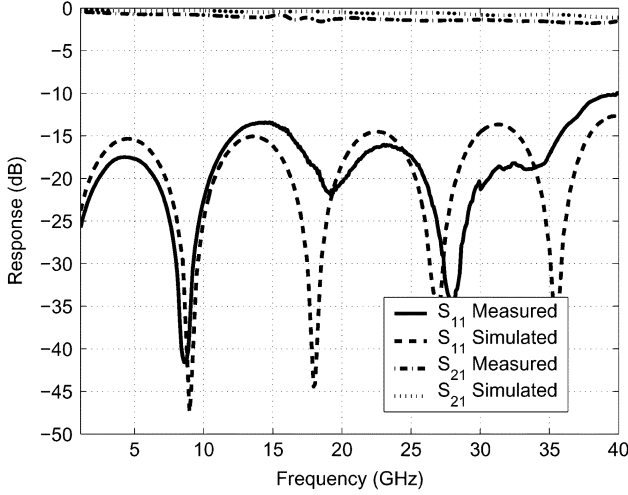


Fig. 5. Measured and simulated S -parameters of the DMTL at 16 V bias.

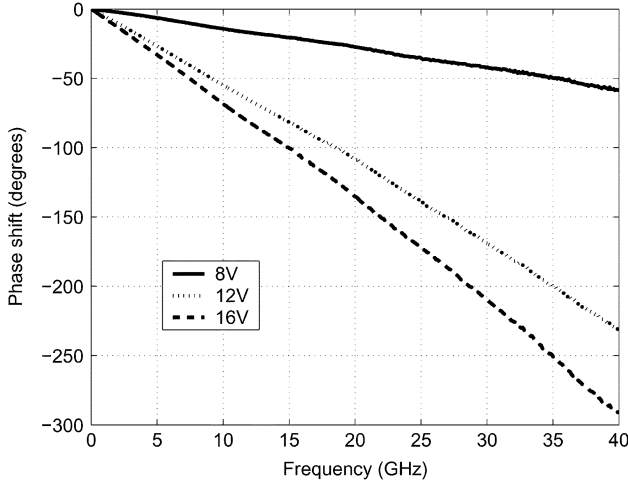


Fig. 6. Total phase shift versus frequency for various voltages.

different over the CPW center conductor and the ground conductors. This allows for a bridge height of 950 nm over the CPW center conductor and oxide, and $2.3 \mu\text{m}$ over the edges of the CPW ground plane and oxide. 500 nm of aluminum alloy was then sputtered and patterned on top of the sacrificial layers to produce the bridges, and finally the structure was released. The final released structure is shown in Fig. 2.

V. MEASUREMENTS

Figs. 4 and 5 show the S_{11} and S_{12} measurements of a typical 5.3-mm-long DMTL phase shifter, at 0 and 16 V, respectively. These measurements indicate excellent loss up to 40 GHz (< 1.8 dB), while maintaining acceptable amounts of input reflection (< -10 dB). Simulations results of the DMTL are also shown, performed using Agilent's Advanced Design Suite (ADS). Calibration is performed using an on-wafer SOLT probe calibration.

Fig. 6 shows the phase shift versus frequency for a 16-V bias voltage, which corresponds to full deflection of the bridges. For a loaded transmission line length of 5.3 mm, this produces 538° of phase shift per cm. This agrees well with (3), which predicts a $540^\circ/\text{cm}$ phase shift. Based on the loss measured at full bridge deflection, this demonstrates phase shifts of $170^\circ/\text{dB}$ of total

TABLE I
MEMS LOADED TRANSMISSION LINE PARAMETERS

Parameter	Unloaded Line	Loaded Line(0V)	Loaded Line(16V)
$Z_0(1\text{GHz})$	93 Ω	67 Ω	44 Ω
$Z_0(40\text{GHz})$	93 Ω	58 Ω	35 Ω
C_t	55 pF/m	55 pF/m	55 pF/m
L_t	490 nH/m	490 nH/m	490 nH/m
C_l	0 fF	18 fF	62 fF
ν_{phase}	1.9×10^8 m/s	1.4×10^8 m/s	0.9×10^8 m/s

TABLE II
DMTL COMPARISON AT 40 GHz

Device	Loss	Phase Shift per cm	Phase Shift per dB
Barker [4]	1.7 dB/cm*	120 $^\circ$ /cm*	70 $^\circ$ /dB*
This Work	2.4 dB/cm	538 $^\circ$ /cm	225 $^\circ$ /dB*

*without reflection losses

loss at 40 GHz and $150^\circ/\text{dB}$ at 30 GHz. The unique fabrication of these MEMS bridges allows for state of the art phase shift per unit length and per unit loss.

The parameters of the device, calculated from the measured data, are presented in Table I. Notably, a 3.4:1 tuning of the effective distributed capacitance is observed. Note that the characteristic impedance is dependant on the frequency, due to the effect of the periodic Bragg reflections.

Table II shows the phase shifter performance comparison with the previously published DMTL phase shifter by Barker [4]. The losses in both cases do not include reflection losses for proper comparison. This comparison indicates a significantly higher figure of merit—phase shift per dB of loss—of the new design.

VI. CONCLUSION

The successful design and fabrication of a high tuning MEMS distributed analog phase shifter has been presented. Measurements of this device indicate that the phase shifter achieves 170° of phase shift per dB loss at 40 GHz, including all losses. This demonstrates the largest figure of merit performance of MEMS analog phase shifters to date. This fabrication process is also suitable to construct stand-alone low-loss lumped RF capacitors.

REFERENCES

- [1] B. Lakshminarayanan and T. Weller, "MEMS phase shifters using cascaded slow-wave structures for improved impedance matching and/or phase shift," in *Proc. IMS'04 Conf.*, Fort Worth, TX, Jun. 2004, pp. 725–728.
- [2] N. S. Barker and G. M. Rebeiz, "Distributed MEMS true-time delay phase shifters and wide-band switches," *IEEE Trans. Microw. Theory Tech.*, vol. 46, no. 11, pp. 1881–1890, Nov. 1998.
- [3] A. Borgioli, Y. Liu, A. Nagra, and R. A. York, "Low-loss distributed MEMS phase shifter," *IEEE Microw. Guided Wave Lett.*, vol. 10, no. 1, pp. 7–9, Jan. 2000.
- [4] N. S. Barker and G. M. Rebeiz, "Optimization of distributed MEMS phase shifters," in *IEEE MTT-S Int. Dig.*, Jun. 1999, pp. 299–302.
- [5] M. J. W. Rodwell *et al.*, "Active and nonlinear wave propagation devices in ultrafast electronics and optoelectronics," *Proc. IEEE*, vol. 83, no. 7, pp. 1037–1059, Jul. 1994.
- [6] G. M. Rebeiz, *RF MEMS Theory, Design, and Technology*. New York: Wiley, 2003.
- [7] J. Zou, C. Liu, J. Schutt-Aine, J. Chen, and S. M. Kang, "Development of wide tuning range MEMS tunable capacitors for wireless communications systems," in *IEDM Tech. Dig.*, Dec. 2000, pp. 403–406.